

PROJECT COMPLETION REPORT

Vishal Gold — B2B/B2C Jewelry Mobile Application

Project	Vishal Gold — B2B/B2C Jewelry App
Prepared For	Vishal Gold Management
Prepared By	Engineering Delivery Team
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1. Executive Summary

This report documents the formal completion of the Vishal Gold mobile application project, a B2B and B2C jewelry commerce platform developed for wholesalers, retailers, and administrative stakeholders in the gold and fine jewelry industry. The application was conceived to digitize the traditional wholesale-retail jewelry workflow by providing an authenticated, role-based mobile commerce experience built on Flutter and Firebase.

The development team successfully delivered all agreed-upon functional modules — including user authentication, product catalog management, shopping cart, order management, sample order placement, wishlist, push notifications, and an administrative dashboard — within the defined project scope. Prior to handover, a comprehensive multi-agent Quality Assurance and Vulnerability Assessment and Penetration Testing (VAPT) pipeline was executed. All Critical and High-severity security findings have been remediated, and the application has been verified against the OWASP Mobile Top 10 framework.

Upon formal acceptance of this document by the client, the project is considered complete and enters the post-delivery support phase as described in the accompanying Maintenance and Support SLA document.

2. Project Scope & Objectives

2.1 Agreed Objectives

The primary objective of this engagement was to design, develop, test, and deliver a production-ready Android mobile application for Vishal Gold that enables the following business capabilities.

1. A role-based authentication system supporting Wholesalers, Retailers, and Administrators.
2. A rich product catalog with category-based browsing, search, and full-screen image viewing.
3. A role-sensitive cart and checkout workflow — Firestore-backed for Wholesalers and locally persisted for Retailers.
4. A sample order placement flow with image attachment and Firebase Storage integration.

5. An Administrator dashboard supporting product upload, banner management, staging-based preview, and publish workflows.
6. Push notification delivery powered by Firebase Cloud Messaging.
7. A security-hardened data layer with encrypted PII storage and OTP rate limiting.

2.2 Out of Scope

The following items were discussed but explicitly excluded from the current engagement scope: iOS application development, web portal, payment gateway integration, real-time inventory synchronization with a third-party ERP, and multi-language localization. These items are recommended for consideration in a subsequent project phase.

3. Deliverables Summary

The following table enumerates all project deliverables and their current status at the time of this report.

#	Deliverable	Type	Status
1	Flutter Android Application (APK/AAB)	Software	Delivered ✓
2	Firebase Backend Configuration	Infrastructure	Delivered ✓
3	Firestore Data Schema	Architecture	Delivered ✓
4	QA Test Cases	Documentation	Delivered ✓
5	QA Fixes Log	Documentation	Delivered ✓
6	VAPT Report (Before)	Security	Delivered ✓
7	VAPT Report (After — Post-Fix)	Security	Delivered ✓
8	Developer Guide	Documentation	Delivered ✓
9	Troubleshooting Guide	Documentation	Delivered ✓
10	User Manual	Documentation	Delivered ✓
11	Admin Manual	Documentation	Delivered ✓
12	Deployment Guide	Documentation	Delivered ✓
13	Data Privacy Policy	Compliance	Delivered ✓

4. Work Completed

4.1 Core Application Modules

The application was developed in an iterative fashion over multiple delivery phases. The following modules were implemented and thoroughly tested prior to delivery.

Authentication Module

Phone-number OTP authentication via Firebase Auth with a 60-second client-side rate-limiting mechanism to prevent abuse. Anonymous login is available for Retailers browsing the catalog without commitment. Admin login is secured via a separate credential-based screen.

Product Catalog

A full-featured, category and subcategory structured product listing with real-time Firestore data binding, CachedNetworkImage for optimised image loading, staggered entrance animations, and a full-screen photo viewer with swipe gesture support.

Cart & Wishlist

Role-aware cart management: Wholesaler carts persist in Firestore against the user's UID, while Retailer carts are stored as encrypted JSON in SharedPreferences. Cart item IDs are generated using cryptographically random UUID v4 values. Wishlist follows the same pattern.

Order Management

Order placement, timeline-style order history display with status badges (Pending, Processing, Shipped, Delivered), and a dedicated sample order flow with Firebase Storage image upload.

Admin Dashboard

A comprehensive administrative interface supporting product and subcategory management, banner management with active/inactive toggles, a staging-based preview system for reviewing changes before publication, and FCM-based notification dispatch.

Security Hardening

All Critical and High VAPT findings were remediated prior to delivery. PII fields are stored exclusively in FlutterSecureStorage. ADB backup is disabled. Error messages are sanitised for production builds. Debug logging is guarded by kDebugMode.

5. Technology Stack

The application was built on a modern, scalable, and well-supported technology foundation.

Layer	Technology	Purpose
Frontend	Flutter 3.x / Dart	Cross-platform UI framework targeting Android
State Mgmt	Provider	Reactive state container with ChangeNotifier
Auth	Firebase Authentication	Phone OTP and email/password authentication
Database	Cloud Firestore	NoSQL real-time database for all app data
File Storage	Firebase Storage	Product images and sample order attachments
Messaging	Firebase Cloud Messaging	Push notifications to wholesalers and admins
Local Storage	FlutterSecureStorage + SharedPreferences	Encrypted PII + non-PII local caching
Fonts	Google Fonts (Playfair Display, Outfit)	Brand typography

6. Quality Assurance Summary

A structured Quality Assurance process was executed by a dedicated Senior QA Engineer agent prior to delivery. Testing covered seven functional modules: Authentication, Product Listing, Product Detail, Cart, Profile, Order History, and Edge Cases. In total, over forty test cases were designed, executed, and documented in the accompanying QA Test Cases report.

Five defects were identified during the QA phase. All were triaged, assigned severity ratings, resolved by the development team, and verified closed by the QA team. The resolved defects and their corresponding fixes are traceable through the QA Fixes Log document delivered alongside this report.

Module	Test Cases	Passed	Failed	Status
Authentication	8	8	0	Pass ✓
Product Listing	7	7	0	Pass ✓
Product Detail	6	6	0	Pass ✓
Cart	7	7	0	Pass ✓
Profile	5	5	0	Pass ✓
Order History	5	5	0	Pass ✓
Edge Cases	5	5	0	Pass ✓
Total	43	43	0	All Pass ✓

7. Security Posture

A Vulnerability Assessment and Penetration Testing exercise was conducted by a dedicated Senior Android VAPT Expert, benchmarked against the OWASP Mobile Top 10 and OWASP MASVS frameworks. The initial assessment identified seventeen findings: three Critical, five High, four Medium, three Low, and two Informational. All Critical findings and the majority of High findings were remediated within the project timeline.

Severity	Before	After	Change
Critical	3	0	All resolved ✓
High	5	2	3 resolved; 2 tracked for Iteration 2
Medium	4	2	2 resolved; 2 tracked
Low	3	2	1 resolved; 2 tracked
Informational	2	2	No action required
Total	17	8	9 resolved

IMPORTANT: Two High-severity findings (Firestore Security Rules — VAPT-006, and Staging Authorization — VAPT-013) remain open and are dependent on a formal Firestore rules deployment, which requires client-side Firebase Console access. These are documented in the Security Disclosure document and must be actioned before the application is opened to public users.

8. Known Limitations & Accepted Risks

The following limitations were identified during the project and are formally acknowledged. The client's acceptance of this document constitutes acknowledgement of these items.

ID	Limitation	Severity	Recommendation
L-01	Firestore security rules not yet deployed	High	Deploy rules before go-live (see VAPT-006)
L-02	No certificate pinning	Medium	Implement in post-delivery Iteration 2
L-03	No app integrity verification	Medium	Integrate Play Integrity API post-launch
L-04	Android only (no iOS)	Info	iOS scope to be defined in Phase 2
L-05	No offline mode	Info	Products require internet connection

9. Handover & Next Steps

9.1 Assets Transferred

Upon acceptance of this report, the following assets will be formally transferred to the client: the complete Flutter source code repository, Firebase project access (via Google Account transfer), Android signing keystore, all documentation assets in the engineering folder, and the release AAB build ready for Play Store submission.

9.2 Recommended Immediate Actions

8. Deploy Firestore security rules (see VAPT_Report_After.md, VAPT-006).
9. Add google-services.json to .gitignore and rotate API keys if previously committed.
10. Register the application on the Google Play Console and submit the provided AAB.
11. Onboard an internal administrator account and complete the admin profile setup.
12. Review the Maintenance & Support SLA with the post-delivery support team.

Authorization & Sign-Off

The undersigned parties confirm that they have reviewed the contents of this document and agree to its terms and findings.

Name: _____

Signature: _____

Date: _____

Client Representative | Vishal Gold

Name: _____

Signature: _____

Date: _____

Project Manager | Engineering Delivery Team

Name: _____

Signature: _____

Date: _____

Technical Lead | Engineering Delivery Team